



C S C I 8314

Fall 2026

SPARSE MATRIX COMPUTATIONS

Class time : MW 2:30 pm – 3:45pm
Room : Keller Hall 3-125
Instructor : Yousef Saad

September 4, 2026

About this class: Objectives

Set 1 An introduction to sparse matrices and sparse matrix computations: Sparse matrices; Sparse matrix direct methods; Graph theory and sparse graphs.

Set 2 Applications of sparse matrix techniques: Graph Laplaceans; Networks; Standard Applications (PDEs, ..); Applications in machine learning (Graph embeddings, Graph Neural Networks)

Set 3 Iterative methods and eigenvalue problems: Iterative methods for linear systems; Algorithms for sparse eigenvalue problems and the SVD

Logistics:

- ▶ We will be using mostly github to post class material.
- ▶ Canvas will be for grades, communications, possibly quizzes etc.
[github site for csci8314](#)
- ▶ Take a look at the syllabus ...
- ▶ The github repository will be updated on a regular basis

About lecture notes:

- ▶ Lecture notes (like this first set) will be posted on github – usually before the lecture.
- ▶ Review them to get some understanding if possible before class.
- ▶ Read the relevant section (s) in the texts or provided references
- ▶ Lecture note sets are grouped by topics rather than by lecture.
- ▶ In the notes the symbol  1 indicates suggested easy exercises or questions – often [not always] done in class.
- ▶ Also: occasional practice exercises posted

About the quizzes

- ▶ 10-15 mn each.
- ▶ Should be quick and easy tests to get you to review the material - not heavy exercises
- ▶ Modality [online or on papers ...] will be decided -

- ▶ Will use either Matlab or Python to quickly illustrate and test algorithms.
- ▶ Scripts will be posted in the 'codes' section on githib.

What I expect you to learn from this course

Part 1

1. Sparse matrices;
 2. Graph representations;
 3. Sparse direct methods for linear systems;
-

Part 2

4. Paths in graphs; Markov Chains;
 5. Graph centrality;
 6. Graph Laplaceans and applications; Clustering;
 7. Graph embedding.
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Part 3

8. Iterative methods for linear systems ;
9. Projection methods and Krylov subspace methods;
10. Eigenvalue problems;

CSCI 8314: SPARSE MATRIX COMPUTATIONS

GENERAL INTRODUCTION

- ▶ General introduction - a little history
- ▶ Motivation
- ▶ Resources
- ▶ What will this course cover
- ▶ Examples of problems leading to sparse matrix computations

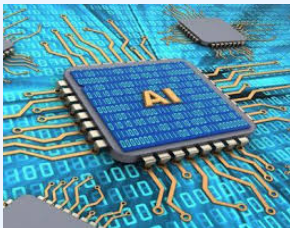
Historical Perspective: Focus of numerical linear algebra

- ▶ **1940s–1950s:** Major issue: flutter problem in aerospace engineering
→ **eigenvalue problem** [cf. Olga Taussky Todd] → LR, QR, .. → ‘EISPACK’
- ▶ **1960s:** Problems related to the power grid promoted what we would call today **general sparse matrix techniques**
- ▶ **1970s–** Automotive, Aerospace, ...: Computational Fluid Dynamics (**CFD**)
- ▶ **Late 1980s:** Thrust on **parallel matrix computations**.
- ▶ **Late 1990s:** Spur of interest in **“financial computing”**
- ▶ **Current:** **Machine Learning, AI**

Solution of PDEs (e.g., Fluid Dynamics) and problems in mechanical eng. (e.g. structures) major force behind numerical linear algebra algorithms in the past few decades.

- ▶ Strong new forces are now reshaping the field today: Applications related to the use of “data”
- ▶ Machine learning is appearing in every application:
 - design of materials, design of drugs
 - machine learning in geophysics
 - self-driving cars, ..
 - .. even in arts (music, film making)

Big impact on the economy



- ◆ Artificial General Intelligence (AGI) is upon us → New economy driven by openAI, Anthropic, Meta, Ali-Baba, Tencent, ...,
- ◆ Huge impact on **Jobs**

- ◆ Big uncertainty in classical areas: Manufacturing; Services; ... what will remain?



Sparse matrices: a brief history

Sparse matrices have been identified as important early on – terminology is quite old. Gauss defined the first method for such systems in 1823. Varga used explicitly the term 'sparse' in his 1962 book on iterative methods.

<https://www-users.cs.umn.edu/~saad/PDF/icerm2018.pdf>

- ▶ Special techniques for sparse problems coming from Partial Differential Equations
- ▶ 1960s: birth of the general methodology available today
- ▶ Graphs introduced as tools for sparse Gaussian elimination in 1961 [Seymour Parter]

- ▶ Early work on reordering for banded systems, envelope methods
- ▶ Various reordering techniques for general sparse matrices introduced.
- ▶ Minimal degree ordering [Markowitz - 1957] ...
- ▶ ... later used in Harwell MA28 code [Duff] - released in 1977.
- ▶ Tinney-Walker Minimal degree ordering for power systems [1967]
- ▶ Nested Dissection [A. George, 1973]
- ▶ SPARSPAK [commercial code, Univ. Waterloo]
- ▶ Elimination trees, symbolic factorization, ...

History: development of iterative methods

- ▶ 1950s up to 1970s : focus on “relaxation” methods
- ▶ Development of ‘modern’ iterative methods took off in the mid-70s. but...
- ▶ ... The main ingredients were in place earlier [late 40s, early 50s: Lanczos; Arnoldi ; Hestenes (a local!) and Stiefel;]
- ▶ The next big advance was the push of ‘preconditioning’: in effect a way of combining iterative and (approximate) direct methods – [Meijerink and Van der Vorst, 1977]

<https://www-users.cs.umn.edu/~saad/PDF/NIST75th.pdf>

History: eigenvalue problems

- ▶ Another parallel branch was followed in sparse techniques for large eigenvalue problems.
- ▶ A big problem in 1950s and 1960s : flutter of airplane wings.. This leads to a large (sparse) eigenvalue problem
- ▶ Overlap between methods for linear systems and eigenvalue problems [Lanczos, Arnoldi]

- ▶ SuiteSparse site (Formerly: Florida collection)

<https://sparse.tamu.edu/about>

- ▶ *scipy.sparse* package from *scipy* –
- ▶ Also: *scikit-sparse* provides wrappers for SuiteSparse library
- ▶ Old: SPARSKIT, etc. [SPARSKIT = old written in Fortran. + more recent ‘solvers’]

<http://www.cs.umn.edu/~saad/software>

Books: on sparse direct methods.

- ▶ Book by Tim Davis [SIAM, 2006] see syllabus for info
- ▶ An old reference [Still a great book]: Alan George and Joseph W-H Liu, **Computer Solution of Large Sparse Positive Definite Systems**, Prentice-Hall, 1981.
- ▶ Of interest, mostly for reference:
 - I. S. Duff and A. M. Erisman and J. K. Reid, **Direct Methods for Sparse Matrices**, Clarendon press, Oxford, 1986; 2nd edition: 2017.
 - Some coverage in Golub and van Loan [John Hopkins, 4th Ed.,]

Overall plan for this course

- ▶ Start: sparse matrices in general, and how to work with them
- ▶ Graph theory viewpoint for sparsity
- ▶ We will then spend some time on sparse direct methods [how much is TBD]
- ▶ .. back to graphs: Graph Laplaceans and applications; Networks; ...
- ▶ .. and then on eigenvalue problems and
- ▶ ... iterative methods for linear systems
- ▶ ... Plan is somewhat dynamic
- ▶ ... at the end of semester: a few lectures given by you [presentations]

Some applications that lead to sparse matrices

- ▶ The classical: CFD, electrical networks,
- ▶ ... and the modern:
 - Graph algorithms and tools (Sparse graphs, graph coarsening, graphs and sparse methods). ..
 - Dimension reduction methods; Graph embeddings;
 - Specific machine learning algorithms; unsupervised/ supervised learning;
 - Network analysis.
 - Deep learning;
 - ...

Example: Fluid flow

Physical Model



Nonlinear PDEs



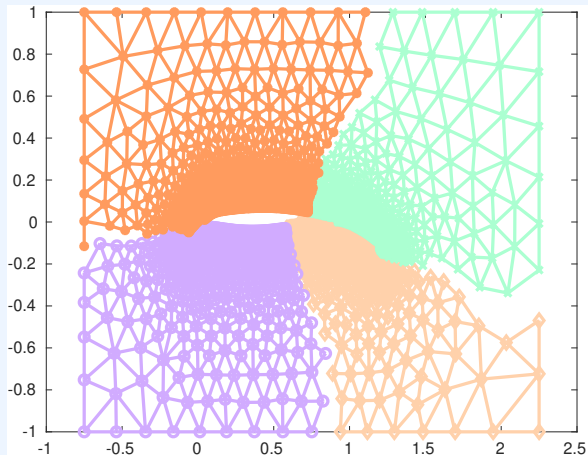
Discretization



Linearization (Newton)

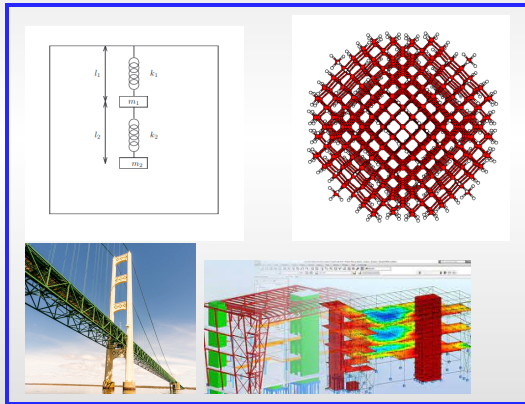


Sparse Linear Systems $Ax = b$



Example: Eigenvalue Problems

- ▶ Many applications require the computation of a few eigenvalues + associated eigenvectors of a matrix A

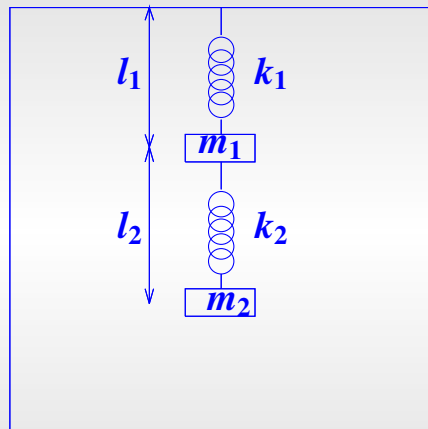


- Structural Engineering – (Goal: frequency response)
- Electronic structure calculations [Schrödinger equation..] – Quantum chemistry
- Stability analysis [e.g., electrical networks, mechanical system,..]
- ...

Example: Vibrations

- ▶ Vibrations in mechanical systems. See:
www.cs.umn.edu/~saad/eig_book_2ndEd.pdf

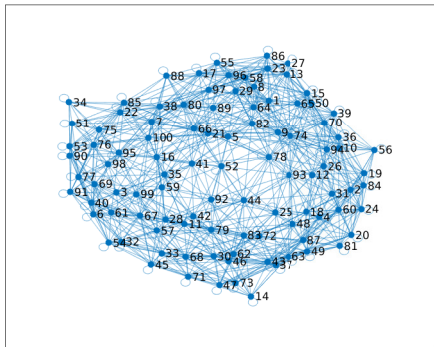
Problem: Determine the vibration modes of the mechanical system [to avoid resonance]. See details in Chapter 10 (sec. 10.2) of above reference.



- ▶ Problem type: Eigenvalue Problem

Example: Google Rank (pagerank)

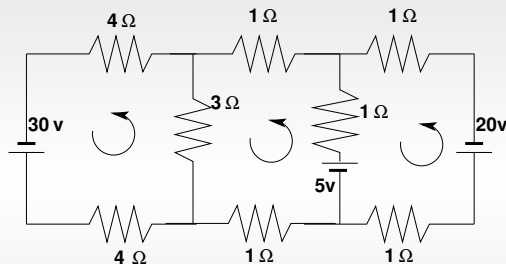
If one were to do a random walk from web page to web page, following each link on a given web page at random with equal likelihood, which are the pages to be encountered this way most often?



► Problem type: (homogeneous) Linear system. Eigenvector problem.

Example: Power networks

- ▶ Electrical circuits .. [Kirchhoff's voltage Law]



Problem: Determine the loop currents in a an electrical circuit - using Kirchhoff's Law ($V = RI$)

- ▶ Problem: Sparse Linear Systems [at the origin Sparse Direct Methods]

Example: Economics/ Marketing/ Social Networks

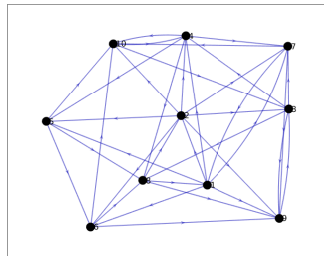
- ▶ Given: an influence graph G : g_{ij} = strength of influence of j over i
- ▶ Goal: charge member i price p_i in order to maximize profit
- ▶ Utility for member i : [x_i = consumption of i]

$$u_i = ax_i - bx_i^2 + \sum_{j \neq i} g_{ij}x_j - p_i x_i$$

- 1: 'Monopolist' fixes prices; 2: agent i fixes consumption x_i

Result: Optimal pricing proportional to **Bonacich** centrality:

$(I - \alpha G)^{-1} \mathbf{1}$ where $\alpha = \frac{1}{2b}$ [Candogan et al., 2012 + many refs.]



- ▶ 'centrality' defines a measure of importance of a node (or an edge) in a graph
- ▶ Many other ideas of centrality in graphs [degree centrality, betweenness centrality, closeness centrality, ...]
- ▶ Important application: Social Network Analysis

Example: Method of least-squares

- ▶ First use of least squares by Gauss, in early 1800's:

A planet follows an elliptical orbit according to $ay^2 + bxy + cx + dy + e = x^2$ in cartesian coordinates. Given a set of noisy observations of (x, y) positions, compute a, b, c, d, e , and use to predict future positions of the planet. This least squares problem is nearly rank-deficient and hence very sensitive to perturbations in the observations.

- ▶ Problem type: Least-Squares system

Read Wikipedia's article on planet ceres:

[http://en.wikipedia.org/wiki/Ceres_\(dwarf_planet\)](http://en.wikipedia.org/wiki/Ceres_(dwarf_planet))

Example: Dynamical systems and epidemiology

A set of variables that fill a vector y are governed by the equation

$$\frac{dy}{dt} = Ay$$

Determine $y(t)$ for $t > 0$, given $y(0)$ [called 'orbit' of y]

- ▶ Problem type: (Linear) system of ordinary differential equations.

Solution:

$$y(t) = e^{tA}y(0)$$

- ▶ Involves exponential of A [think Taylor series], i.e., a **matrix function**

- ▶ This is the simplest form of dynamical systems (linear).
- ▶ Consider the slightly more complex system:

$$\frac{dy}{dt} = A(y)y$$

- ▶ Nonlinear. Requires ‘integration scheme’.

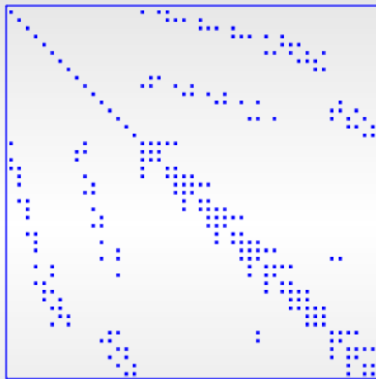
General Problems in Numerical Linear Algebra (dense & sparse)

- ▶ Linear systems: $Ax = b$. Often: A is large and sparse
- ▶ Least-squares problems $\min \|b - Ax\|_2$
- ▶ Eigenvalue problem $Ax = \lambda x$. Several variations -
- ▶ SVD and Low-rank approximation
- ▶ Tensors and low-rank tensor approximation
- ▶ Matrix equations: Sylvester, Lyapunov, Riccati, ..
- ▶ Nonlinear equations – acceleration methods
- ▶ Matrix functions and applications
- ▶ Many many more ...

SPARSE MATRICES

- ▶ Introduction to sparse matrices
- ▶ See Chap. 3 of text
- ▶ See also the various sparse matrix sites
- ▶ Sparse matrices in matlab

What are sparse matrices?



Pattern of a small sparse matrix

- ▶ Vague definition: matrix with few nonzero entries
- ▶ For all practical purposes: an $m \times n$ matrix is sparse if it has $O(\min(m, n))$ nonzero entries.
- ▶ This means roughly a constant number of nonzero entries per row and column -
- ▶ This definition excludes a large class of matrices that have $O(\log(n))$ nonzero entries per row.
- ▶ Other definitions use a slow growth of nonzero entries with respect to n or m .

“..matrices that allow special techniques to take advantage of the large number of zero elements.” (J. Wilkinson)

A few applications which lead to sparse matrices:

Structural Engineering, Computational Fluid Dynamics, Reservoir simulation, Electrical Networks, optimization, Google Page rank, information retrieval (LSI), circuit simulation, device simulation,

Goal of Sparse Matrix Techniques

- ▶ To perform standard matrix computations economically i.e., without storing the zeros of the matrix.

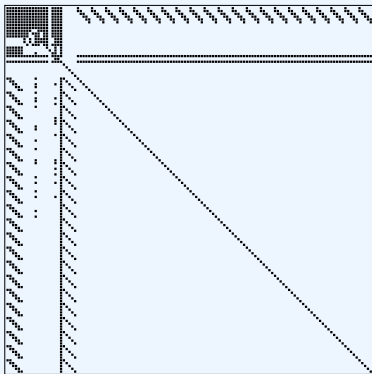
Example: To add two square dense matrices of size n requires $O(n^2)$ operations. To add two sparse matrices A and B requires $O(nnz(A) + nnz(B))$ where $nnz(X)$ = number of nonzero elements of a matrix X .

- ▶ For typical Finite Element /Finite difference matrices, number of nonzero elements is $O(n)$.

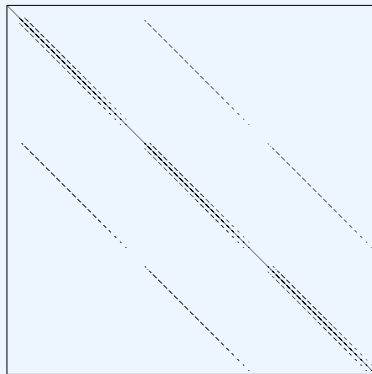
Remark: A^{-1} is usually dense, but L and U in the LU factorization may be reasonably sparse (if a good technique is used)

-  **2** Look up Cayley-Hamilton's theorem if you do not know about it.
-  **3** Show that the inverse of a matrix (when it exists) can be expressed as a polynomial of A , where the polynomial is of degree $\leq n - 1$.
-  **4** When is the degree $< n - 1$? [Hint: look-up minimal polynomial of a matrix]
-  **5** What is the pattern of the inverse of a tridiagonal matrix? a bidiagonal matrix?

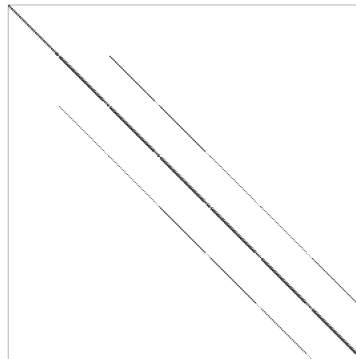
Nonzero patterns of a few sparse matrices



ARC130: Unsymmetric matrix from laser problem. a.r.curtis, oct 1974



SHERMAN5: fully implicit black oil simulator 16 by 23 by 3 grid, 3 unk

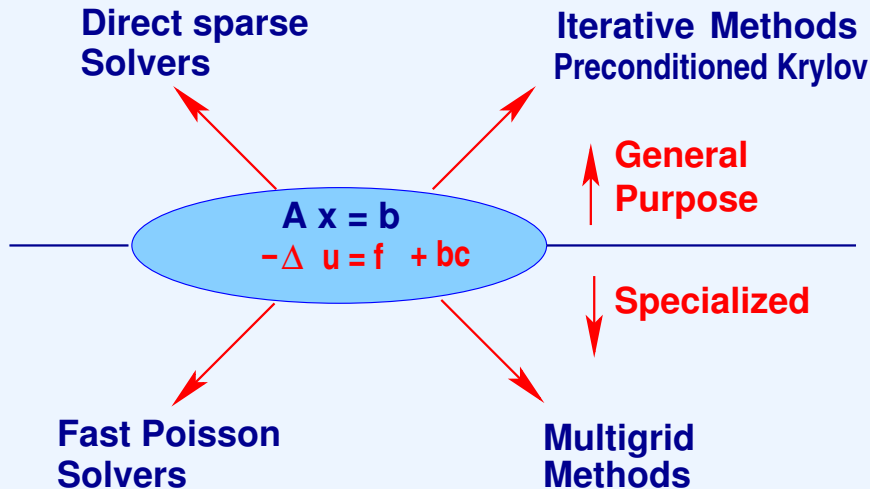


PORES3: Unsymmetric MATRIX FROM PORES

Types of sparse matrices

- ▶ Two types of matrices: **structured** and **unstructured**
- ▶ The matrices PORES3 and SHERMAN5 are from Oil Reservoir Simulation. Often: 3 unknowns per mesh point (Oil , Water saturations, Pressure). Structured matrices.
- ▶ 40 years ago reservoir simulators used rectangular grids.
- ▶ Modern simulators: Finer, more complex physics $\Rightarrow$ harder + larger systems. Also: unstructured matrices
- ▶ A naive but representative challenge problem: $100 \times 100 \times 100$ grid + about 10 unknowns per grid point $\Rightarrow N \approx 10^7$, and $nnz \approx 7 \times 10^8$.

Solving sparse linear systems: existing methods



Two types of methods for general systems:

- ▶ Direct methods : based on sparse Gaussian elimination, sparse Cholesky,...
- ▶ Iterative methods: compute a sequence of iterates which converge to the solution - preconditioned Krylov methods..

Remark: These two classes of methods have always been in competition.

- ▶ 40 years ago solving a system with $n = 10,000$ was a challenge
- ▶ Now you can solve this in a fraction of a second on a laptop.

- ▶ Sparse direct methods made huge gains in efficiency. As a result they are very competitive for 2-D problems.
- ▶ 3-D problems lead to more challenging systems [inherent to the underlying graph]

Difficulty:

- ▶ No robust 'black-box' iterative solvers.
- ▶ At issue: Robustness in conflict with efficiency.
- ▶ Iterative methods are starting to use some of the tools of direct solvers to gain 'robustness'

Consensus:

1. Direct solvers are often preferred for two-dimensional problems (robust and not too expensive).
2. Direct methods loose ground to iterative techniques for three-dimensional problems, and problems with a large degree of freedom per grid point,

Sparse matrices in matlab

▶ Matlab supports sparse matrices to some extent.

▶ Can define sparse objects by conversion

```
A = sparse(X) ; X = full(A)
```

▶ Can show pattern

```
spy(X)
```

▶ Define the analogues of `ones`, `eye`:

```
speye(n,m) ; spones(pattern)
```

- ✦ A few reordering functions provided.. [will be studied in detail later]

`symrcm, symamd, colamd, colperm`

- ✦ Random sparse matrix generator: `sprand(S)` or `sprand(m,n, density)`

- ✦ Diagonal extractor-generator utility: `spdiags(A)` , `spdiags(B,d,m,n)`

- ✦ Other important functions: `spalloc(..)` , `find(..)`

Graph Representations of Sparse Matrices

- ✦ Graph theory is a fundamental tool in sparse matrix techniques.

Definition: A graph G is defined as a pair of sets $G = (V, E)$ with $E \subset V \times V$. So G represents a binary relation. The graph is **undirected** if the binary relation is reflexive. It is **directed** otherwise. V is the vertex set and E is the edge set.

Example: Given the numbers 5, 3, 9, 15, 16, show the two graphs representing the relations

R1: Either $x < y$ or y divides x .

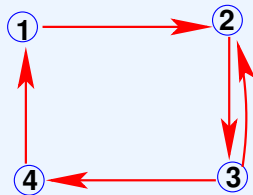
R2: x and y are congruent modulo 3. [$\text{mod}(x,3) = \text{mod}(y,3)$]

- ▶ **Adjacency Graph** $G = (V, E)$ of an $n \times n$ matrix A :
 - Vertices $V = \{1, 2, \dots, n\}$.
 - Edges $E = \{(i, j) | a_{ij} \neq 0\}$.
- ▶ Often self-loops (i, i) are not represented [because they are always there]
- ▶ Graph is **undirected** if the matrix has a symmetric structure:

$$a_{ij} \neq 0 \quad \text{iff} \quad a_{ji} \neq 0.$$

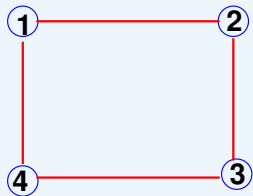
Example: (directed graph)

	★		
		★	
	★		★
★			



Example: (undirected graph)

	★		★
★		★	
	★		★
★		★	



 6 Graph of a tridiagonal matrix? Of a dense matrix?

 7 Adjacency graph of:

$$A = \begin{bmatrix} \star & \star & & & \star & \\ \star & \star & \star & & & \star \\ & \star & \star & & & \\ & & & \star & \star & \\ \star & & & \star & \star & \star \\ & \star & & & \star & \star \end{bmatrix} ?$$

 8 Define: star graph. Show a matrix whose graph is a star graph. Consider 2 situations: Case when center node is labeled first and case when it is labeled last.

- ▶ Note: Matlab has a `graph` function.
- ▶ `G = graph(A)` creates adjacency graph from A
- ▶ G is a matlab class
- ▶ `G.Nodes` will show the vertices of G
- ▶ `G.Edges` will show its edges.
- ▶ `plot(G)` will show a representation of the graph

**9**

Do the following:

- ▶ Load the matrix 'Bmat.mat' located in the class web-site (see 'matlab' folder)
- ▶ Visualize pattern (`spy(B)`) + find: Number of nonzero elements, size, ...
- ▶ Generate graph - omit self-edges: `G = graph(B, 'OmitSelfLoops')`
- ▶ Plot the graph –
- ▶ \$1M question: Any idea on how this plot is generated?